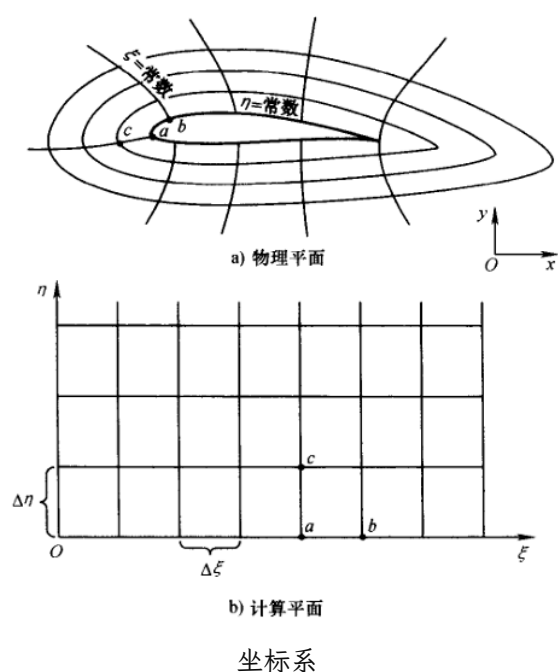




网格生成与坐标变换

计算流体力学

坐标变换



图中为严格围绕着翼型的、非均匀的曲线网格，通过定义新的坐标线 ξ 和 η ，使得翼型表面成为一条坐标线 ($\eta = \text{常数}$)，这种坐标系称为贴近坐标系，网格点自然而然的落在翼型表面上，在物理区域中，网格既不是矩形的，也不是均匀分布的，通常的差分很难应用，所以必须将物理区域内的曲线网格转换成用 ξ 和 η 表示的矩形网格。

将物理平面中的自变量 (x, y, t) 变换成计算平面中的一组新自变量 (ξ, η, τ) ：

$$\xi = \xi(x, y, t)$$

$$\eta = \eta(x, y, t)$$

$$\tau = \tau(t)$$

链式法则给出偏导数：

- 一阶偏导数

$$\frac{\partial}{\partial x} = \left(\frac{\partial}{\partial \xi} \right) \left(\frac{\partial \xi}{\partial x} \right) + \left(\frac{\partial}{\partial \eta} \right) \left(\frac{\partial \eta}{\partial x} \right)$$

$$\frac{\partial}{\partial y} = \left(\frac{\partial}{\partial \xi} \right) \left(\frac{\partial \xi}{\partial y} \right) + \left(\frac{\partial}{\partial \eta} \right) \left(\frac{\partial \eta}{\partial y} \right)$$

$$\frac{\partial}{\partial t} = \left(\frac{\partial}{\partial \xi} \right) \left(\frac{\partial \xi}{\partial t} \right) + \left(\frac{\partial}{\partial \eta} \right) \left(\frac{\partial \eta}{\partial t} \right) + \left(\frac{\partial}{\partial \tau} \right) \left(\frac{d\tau}{dt} \right)$$

- 二阶偏导数

$$\begin{aligned} \frac{\partial^2}{\partial x^2} &= \left(\frac{\partial}{\partial \xi} \right) \left(\frac{\partial^2 \xi}{\partial x^2} \right) + \left(\frac{\partial}{\partial \eta} \right) \left(\frac{\partial^2 \eta}{\partial x^2} \right) + \left(\frac{\partial^2}{\partial \xi^2} \right) \left(\frac{\partial \xi}{\partial x} \right)^2 \\ &\quad + \left(\frac{\partial^2}{\partial \eta^2} \right) \left(\frac{\partial \eta}{\partial x} \right)^2 + 2 \left(\frac{\partial^2}{\partial \eta \partial \xi} \right) \left(\frac{\partial \eta}{\partial x} \right) \left(\frac{\partial \xi}{\partial x} \right) \end{aligned}$$

$$\begin{aligned} \frac{\partial^2}{\partial y^2} &= \left(\frac{\partial}{\partial \xi} \right) \left(\frac{\partial^2 \xi}{\partial y^2} \right) + \left(\frac{\partial}{\partial \eta} \right) \left(\frac{\partial^2 \eta}{\partial y^2} \right) + \left(\frac{\partial^2}{\partial \xi^2} \right) \left(\frac{\partial \xi}{\partial y} \right)^2 \\ &\quad + \left(\frac{\partial^2}{\partial \eta^2} \right) \left(\frac{\partial \eta}{\partial y} \right)^2 + 2 \left(\frac{\partial^2}{\partial \eta \partial \xi} \right) \left(\frac{\partial \eta}{\partial y} \right) \left(\frac{\partial \xi}{\partial y} \right) \end{aligned}$$

$$\begin{aligned} \frac{\partial^2}{\partial x \partial y} &= \left(\frac{\partial}{\partial \xi} \right) \left(\frac{\partial^2 \xi}{\partial x \partial y} \right) + \left(\frac{\partial}{\partial \eta} \right) \left(\frac{\partial^2 \eta}{\partial x \partial y} \right) + \left(\frac{\partial^2}{\partial \xi^2} \right) \left(\frac{\partial \xi}{\partial x} \right) \left(\frac{\partial \xi}{\partial y} \right) \\ &\quad + \left(\frac{\partial^2}{\partial \eta^2} \right) \left(\frac{\partial \eta}{\partial x} \right) \left(\frac{\partial \eta}{\partial y} \right) + \left(\frac{\partial^2}{\partial \eta \partial \xi} \right) \left[\left(\frac{\partial \eta}{\partial x} \right) \left(\frac{\partial \xi}{\partial y} \right) + \left(\frac{\partial \xi}{\partial x} \right) \left(\frac{\partial \eta}{\partial y} \right) \right] \end{aligned}$$



在上述各式中，涉及到网格几何性质的项，如 $\frac{\partial \xi}{\partial x}$, $\frac{\partial \xi}{\partial y}$, $\frac{\partial \eta}{\partial x}$, $\frac{\partial \eta}{\partial y}$ ，称为度量。也要用有限差分计算度量。利用如下逆变换：

$$\begin{aligned} x &= x(\xi, \eta, \tau) \\ y &= y(\xi, \eta, \tau) \\ t &= t(\tau) \end{aligned}$$

雅可比行列式：

$$J \equiv \frac{\partial(x, y)}{\partial(\xi, \eta)} \equiv \begin{vmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{vmatrix}$$

逆变换：

$$\begin{aligned}\frac{\partial}{\partial x} &= \frac{1}{J} \left[\left(\frac{\partial}{\partial \xi} \right) \left(\frac{\partial y}{\partial \eta} \right) - \left(\frac{\partial}{\partial \eta} \right) \left(\frac{\partial y}{\partial \xi} \right) \right] \\ \frac{\partial}{\partial y} &= \frac{1}{J} \left[\left(\frac{\partial}{\partial \eta} \right) \left(\frac{\partial x}{\partial \xi} \right) - \left(\frac{\partial}{\partial \xi} \right) \left(\frac{\partial x}{\partial \eta} \right) \right]\end{aligned}$$

逆度量表示的直接度量：

$$\begin{aligned}\frac{\partial \xi}{\partial x} &= \frac{1}{J} \frac{\partial y}{\partial \eta} \\ \frac{\partial \eta}{\partial x} &= -\frac{1}{J} \frac{\partial y}{\partial \xi} \\ \frac{\partial \xi}{\partial y} &= -\frac{1}{J} \frac{\partial x}{\partial \eta} \\ \frac{\partial \eta}{\partial y} &= \frac{1}{J} \frac{\partial x}{\partial \xi}\end{aligned}$$



物理平面中流动控制的强守恒形式：

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0$$

我们希望在计算平面仍能有如下强守恒形式：

$$\frac{\partial U_1}{\partial t} + \frac{\partial F_1}{\partial \xi} + \frac{\partial G_1}{\partial \eta} = 0$$

对原有的强守恒形式中的空间变量进行变换：

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial \xi} \left(\frac{\partial \xi}{\partial x} \right) + \frac{\partial F}{\partial \eta} \left(\frac{\partial \eta}{\partial x} \right) + \frac{\partial G}{\partial \xi} \left(\frac{\partial \xi}{\partial y} \right) + \frac{\partial G}{\partial \eta} \left(\frac{\partial \eta}{\partial y} \right) = 0$$

乘以 Jacobi 行列式

$$J \frac{\partial U}{\partial t} + J \frac{\partial F}{\partial \xi} \left(\frac{\partial \xi}{\partial x} \right) + J \frac{\partial F}{\partial \eta} \left(\frac{\partial \eta}{\partial x} \right) + J \frac{\partial G}{\partial \xi} \left(\frac{\partial \xi}{\partial y} \right) + J \frac{\partial G}{\partial \eta} \left(\frac{\partial \eta}{\partial y} \right) = 0$$

先考虑如下导数展开

$$\frac{\partial}{\partial \xi} \left[J F \frac{\partial \xi}{\partial x} \right] = J \left(\frac{\partial \xi}{\partial x} \right) \frac{\partial F}{\partial \xi} + F \frac{\partial}{\partial \xi} \left(J \frac{\partial \xi}{\partial x} \right)$$

将右边第二项移到左边得：

$$J \left(\frac{\partial F}{\partial \xi} \right) \left(\frac{\partial \xi}{\partial x} \right) = \frac{\partial}{\partial \xi} \left[J F \frac{\partial \xi}{\partial x} \right] - F \frac{\partial}{\partial \xi} \left(J \frac{\partial \xi}{\partial x} \right)$$

依次类推可以得到如下表达式：

$$\begin{aligned} J \left(\frac{\partial F}{\partial \eta} \right) \left(\frac{\partial \eta}{\partial x} \right) &= \frac{\partial}{\partial \eta} \left[JF \frac{\partial \eta}{\partial x} \right] - F \frac{\partial}{\partial \eta} \left(J \frac{\partial \eta}{\partial x} \right) \\ J \left(\frac{\partial G}{\partial \xi} \right) \left(\frac{\partial \xi}{\partial y} \right) &= \frac{\partial}{\partial \xi} \left[JG \frac{\partial \xi}{\partial y} \right] - G \frac{\partial}{\partial \xi} \left(J \frac{\partial \xi}{\partial y} \right) \\ J \left(\frac{\partial G}{\partial \eta} \right) \left(\frac{\partial \eta}{\partial y} \right) &= \frac{\partial}{\partial \eta} \left[JG \frac{\partial \eta}{\partial y} \right] - G \frac{\partial}{\partial \eta} \left(J \frac{\partial \eta}{\partial y} \right) \end{aligned}$$

将上述表达式带入可得：

$$\begin{aligned} J \frac{\partial U}{\partial t} + \frac{\partial}{\partial \xi} \left(JF \frac{\partial \xi}{\partial x} + JG \frac{\partial \xi}{\partial y} \right) + \frac{\partial}{\partial \eta} \left(JF \frac{\partial \eta}{\partial x} + JG \frac{\partial \eta}{\partial y} \right) - \\ F \left[\frac{\partial}{\partial \xi} \left(J \frac{\partial \xi}{\partial x} \right) + \frac{\partial}{\partial \eta} \left(J \frac{\partial \eta}{\partial x} \right) \right] - G \left[\frac{\partial}{\partial \xi} \left(J \frac{\partial \xi}{\partial y} \right) + \frac{\partial}{\partial \eta} \left(J \frac{\partial \eta}{\partial y} \right) \right] = 0 \end{aligned}$$

取第一个方括号展开可得

$$\frac{\partial}{\partial \xi} \left(J \frac{\partial \xi}{\partial x} \right) + \frac{\partial}{\partial \eta} \left(J \frac{\partial \eta}{\partial x} \right) = \frac{\partial}{\partial \xi} \left(\frac{\partial y}{\partial \eta} \right) - \frac{\partial}{\partial \eta} \left(\frac{\partial y}{\partial \xi} \right) = \frac{\partial^2 y}{\partial \xi \partial \eta} - \frac{\partial^2 y}{\partial \xi \partial \eta} = 0$$

同理可得第二项方括号为零，因此

$$J \frac{\partial U}{\partial t} + \frac{\partial}{\partial \xi} \left(JF \frac{\partial \xi}{\partial x} + JG \frac{\partial \xi}{\partial y} \right) + \frac{\partial}{\partial \eta} \left(JF \frac{\partial \eta}{\partial x} + JG \frac{\partial \eta}{\partial y} \right) = 0$$

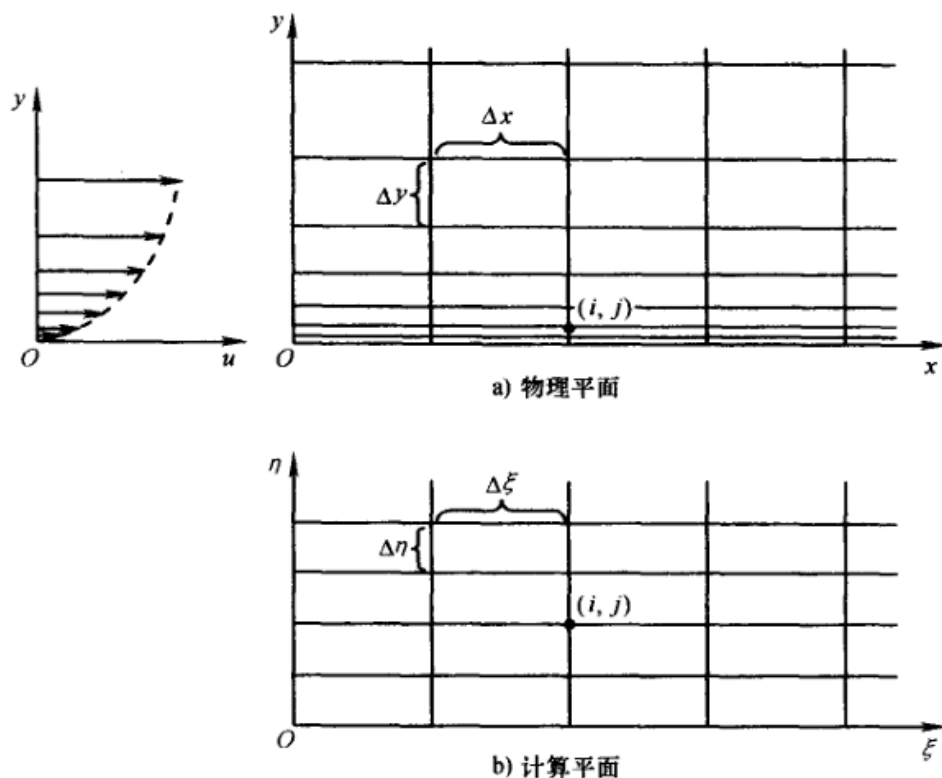
写作

$$\frac{\partial U_1}{\partial t} + \frac{\partial F_1}{\partial \xi} + \frac{\partial G_1}{\partial \eta} = 0$$

这样子我们就得到了一个计算平面的强守恒形式

网格生成

拉伸网格



考虑流过平板的粘性流，在平板表面，速度迅速变化。在垂直方向，水平网格线的分布越靠近物面应该越密。考虑如下简单变换：

$$\begin{aligned}\xi &= x \\ \eta &= \ln(y + 1)\end{aligned}$$

对应逆变换：

$$\begin{aligned}x &= \xi \\ y &= e^\eta - 1\end{aligned}$$

Δy 满足

$$\Delta y = e^\eta \Delta \eta$$

当沿着垂直方向远离物面时，虽然在计算平面是均匀网格，但是得到了逐渐增大的 Δy 值，也就是说，物理平面中的网格沿垂直方向被拉伸了，这就是拉伸网格的含义。



Grids for the supersonic viscous flow over a blunt base

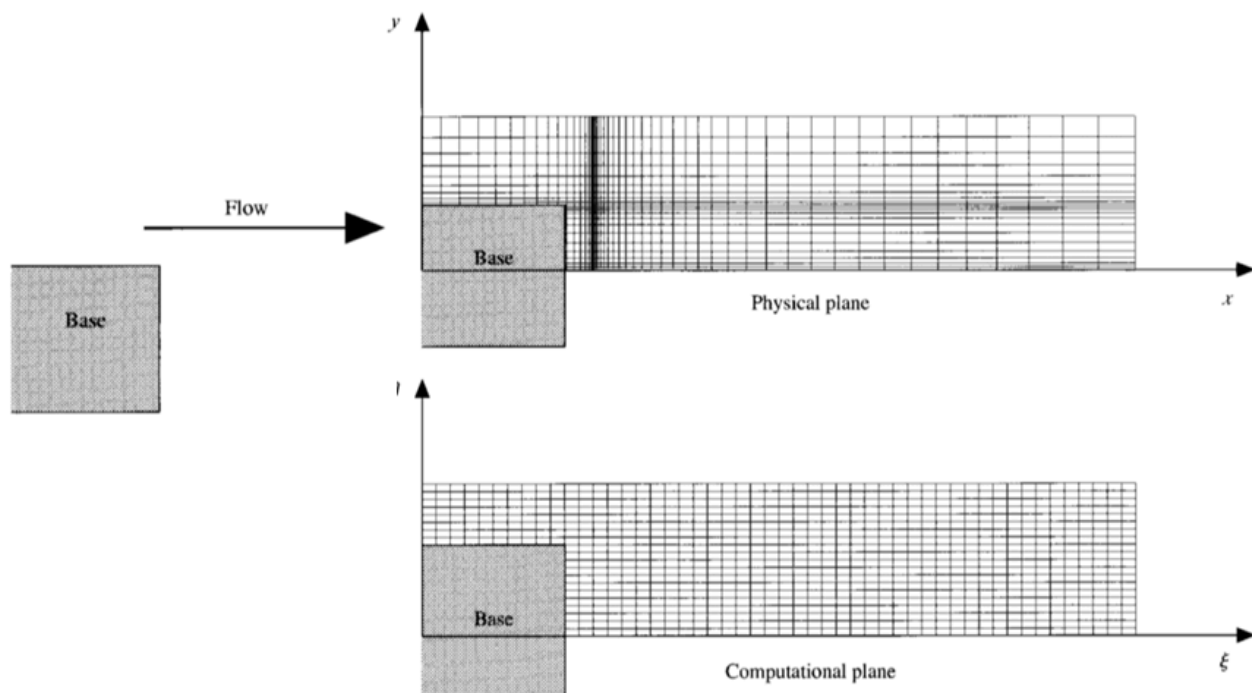


Fig. 5.5 Comparison of uniform and compressed grids.

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研究绕钝体底部的超声速粘性流，此时网格在 x 方向和 y 方向都要进行拉伸。考虑如下形式变换：

$$x = \frac{\xi_0}{A} \{ \sinh[(\xi - x_0)\beta_x] + A \}$$

先进行 x 方向的拉伸，其中

$$A = \sinh(\beta_x x_0)$$

$$x_0 = \frac{1}{2\beta_x} \ln \frac{1 + (e^{\beta_x} - 1)\xi_0}{1 + (e^{-\beta_x} - 1)\xi_0}$$

ξ_0 是物理平面发生最大聚集的点在计算平面中相应的位置， β_x 为常数，控制向 ξ_0 点聚集的程度， β_x 越大，聚集得越密。

网格沿 y 方向的横向拉伸，需要把物理平面分成：底部后面和底部得上方上下两部分，利用变换：

$$y = \frac{(\beta_y + 1) - (\beta_y - 1)e^{-c(\eta-1-\alpha)/(1-\alpha)}}{(2\alpha + 1)(1 + e^{-c(\eta-1-\alpha)/(1-\alpha)})}$$

进行 y 方向的拉伸，其中

$$c = \log \frac{\beta_y + 1}{\beta - 1}$$

我们还可以利用线性插值做拉伸变换。

贴体坐标系

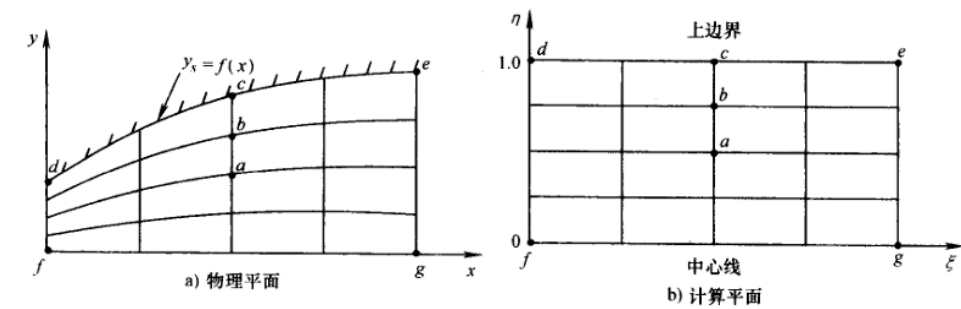


图 5-6 简单的贴体坐标系

在管道流动中，我们希望将边界和中心线都变成坐标线，使得曲线网格完全贴着边界，同时曲线网格也需要变换成计算平面中的矩形网格。这隐含着贴体坐标系的思想。也就是说，我们希望求解区域边界与坐标系得坐标轴一一相符。

但是很多时候我们想要生成的贴体网格并不会像上面的那么简单，它可能是下面这种。一条坐标线与翼型的表面重合，这条坐标线是网格的内边界，用 Γ_1 表示，形状与位置由翼型决定；用 Γ_2 表示网格的外边界，形状和位置是任意的。

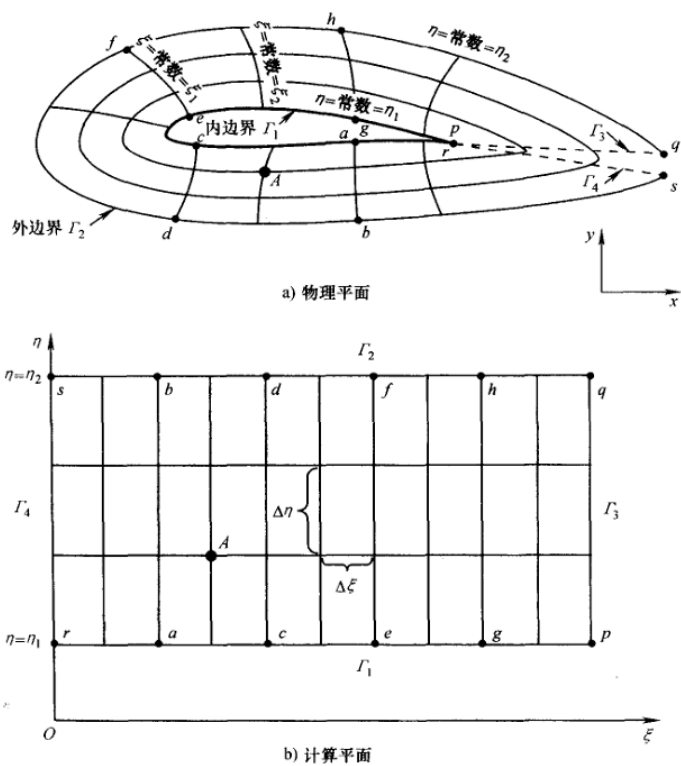


图 5-7 椭圆型方法生成的贴体网格

在物理平面中，沿着内边界 Γ_1 物理坐标已知，也就是说，对于 Γ_1 上任意给定的点，由两个数是已知，即该点的 x 坐标和 y 坐标。同样外边界的物理坐标，于是这启发我们这种变换可以由一个椭圆型偏微分方程确定。考虑如下 Laplace 方程

$$\frac{\partial^2 \xi}{\partial x^2} + \frac{\partial^2 \xi}{\partial y^2} = 0$$

$$\frac{\partial^2 \eta}{\partial x^2} + \frac{\partial^2 \eta}{\partial y^2} = 0$$

逆方程

$$\alpha \frac{\partial^2 y}{\partial \xi^2} - 2\beta \frac{\partial^2 y}{\partial \xi \partial \eta} + \gamma \frac{\partial^2 y}{\partial \eta^2} = 0$$

其中

$$\alpha = \left(\frac{\partial x}{\partial \eta} \right)^2 + \left(\frac{\partial y}{\partial \eta} \right)^2$$

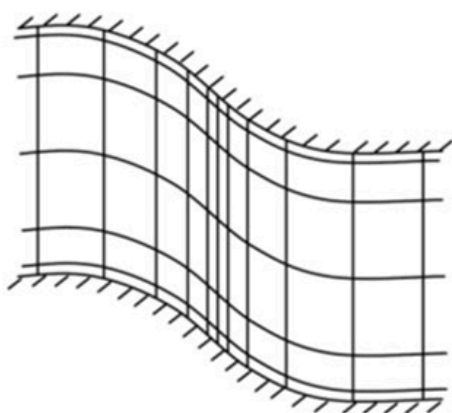
$$\beta = \left(\frac{\partial x}{\partial \xi} \right) \left(\frac{\partial x}{\partial \eta} \right) + \left(\frac{\partial y}{\partial \xi} \right) \left(\frac{\partial y}{\partial \eta} \right)$$

$$\gamma = \left(\frac{\partial x}{\partial \xi} \right)^2 + \left(\frac{\partial y}{\partial \xi} \right)^2$$

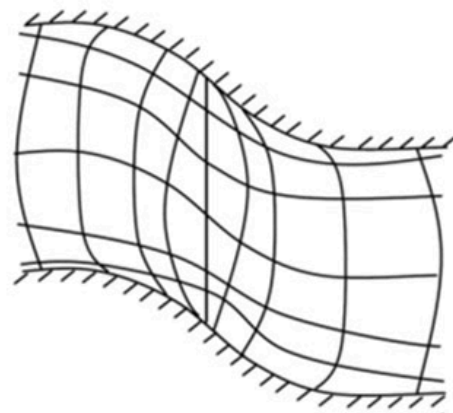
当然，这个方法有一个不足：内部区域的网格分布不易控制；无法做到指定区域网格加密；无法保证网格正交



不足：内部区域的网格分布不易控制
无法做到指定区域网格加密
无法保证网格正交



(Intended Clustered grid)



(Laplace solution)

比如说我们想要达到左边的网格划分效果，这看起来非常棒，因为在边界时密集的，在管道变换的地方也是密集的，但是Laplace的解却是右边的情况，这看起来非常的丑陋。

一个解决办法是将Laplace方程变成Possian方程，也就是添加源项。

我们要求物理空间必须是单连通域，如果是多联通的，可通过切割形成单连通域。

⚠ Warning

椭圆型网格生成方法只是通过求解椭圆型方程来获得区域内部的网格点。网格与方程是两回事。

自适应网格

自适应网格是能够自动向流场中大梯度区域聚集的网格，它利用求解的流场特征确定网格点在物理平面中的位置。自适应网格可以想象成是一种随时间变换的网格，在流场控制方程的时间推进求解中，网格的调整与按时间步计算流场变量的过程同步，在这种求解过程中，流场中大的梯度随时间发展，物理平面中的网格点通过移动去“迎合”它。

一个例子是 Corda 求解绕后台阶的粘性超声速流时所用的网格，其中的坐标变换为

$$\Delta x = \frac{B\Delta\xi}{1 + b(\partial g/\partial x)}$$
$$\Delta y = \frac{C\Delta\eta}{1 + c(\partial g/\partial y)}$$

g 是流场中的原始变量， $\Delta\xi$ 和 $\Delta\eta$ 为常量， b 和 c 是改变物理平面网格间距是用来增加或降低梯度的 B 和 C 是比例因子。

t 时刻， N 点的 x 坐标满足

$$x_N^t = \sum_1^N (\Delta x)_i^t$$

$t + \Delta t$ 时刻， x 坐标为

$$x_N^{t+\Delta t} = \sum_1^N (\Delta x)_i^{t+\Delta t}$$

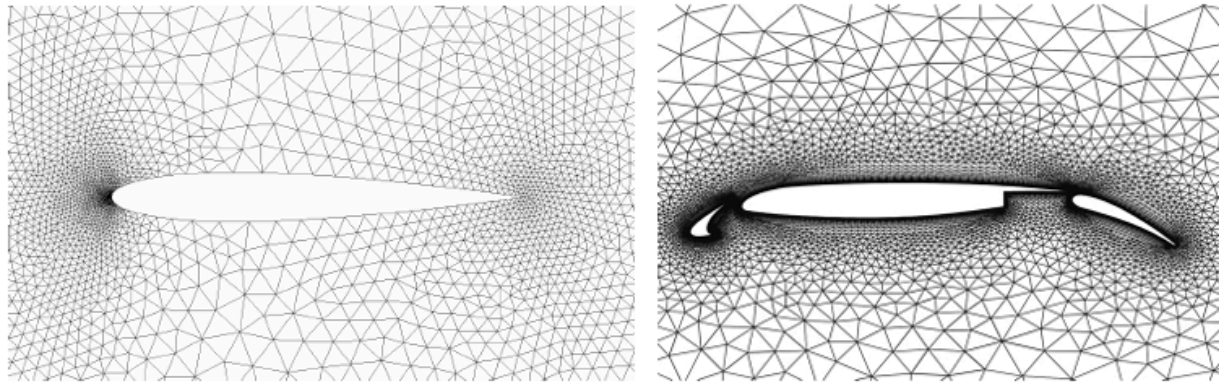
所以

$$\frac{\partial x}{\partial t} \approx \frac{\sum_1^N (\Delta x)_i^{t+\Delta t} - \sum_1^N (\Delta x)_i^t}{\Delta t}$$

非结构化网格

知 [计算几何第五周：完美三角剖分 \(Delaunay Triangulation\)](#) - 知乎

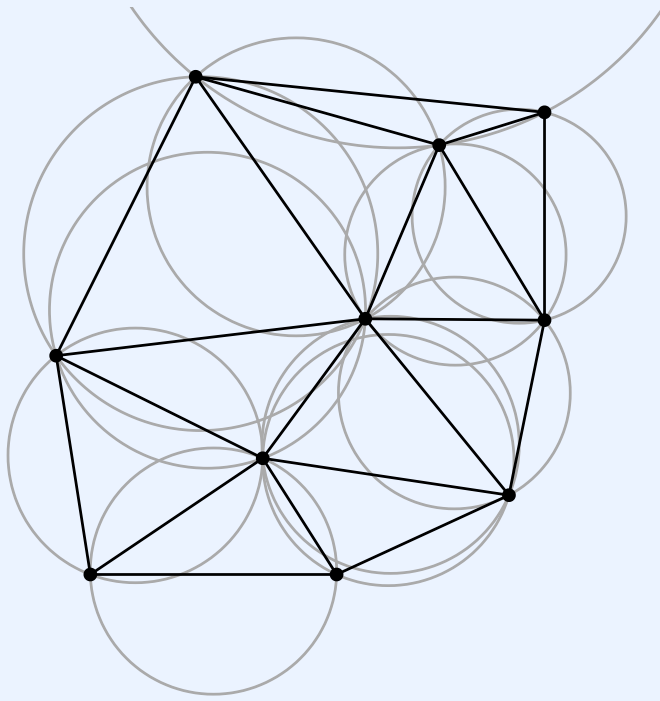
W [Delaunay triangulation](#)



Nodes and grid cells are quasi randomly ordered, i.e., neighbouring cells or grid points cannot be directly identified by their indices.

Delaunay triangulation is a particular way to connect a set of points in a plane to form non-overlapping triangles.

- Incremental
- Divide-and-conquer
- Sweepline



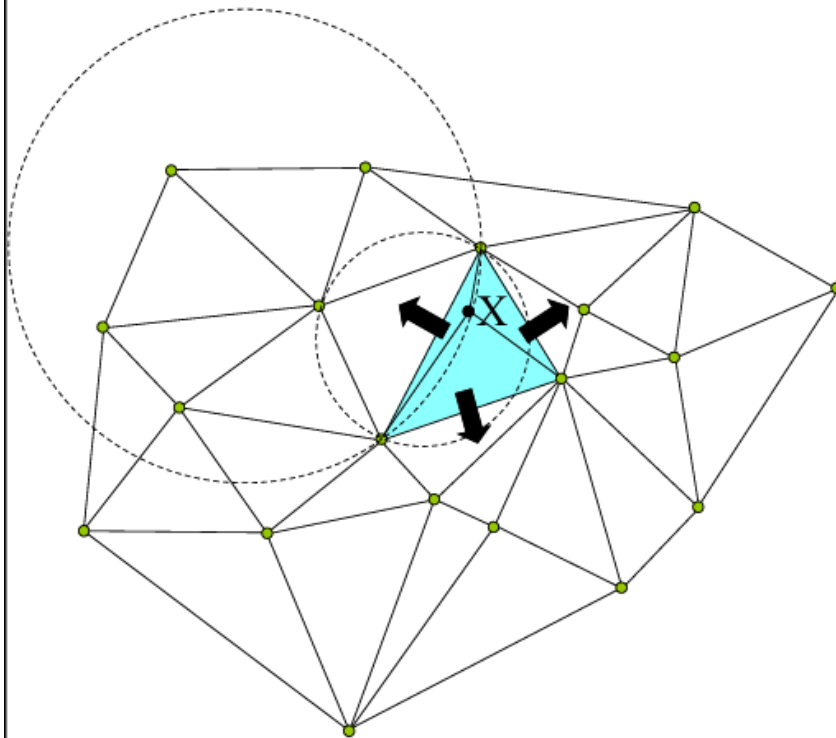
By Gjacquenot

平面中的一组 Delaunay 三角剖分将 $\mathbb{W}$ [凸包](#) 细分为三角形，这些三角形的外接圆不包含其他任何点，也就是说，每个外接圆的生成点都在其圆周上，但集合中的所有其他点都在它之外。Delaunay 三角剖分最大化了三角剖分中三角形的最小角，换句话说，此算法尽量避免出现“极瘦”的三角形。

补充 Voronoi 图

Voronoi 图由一组由连接两邻点直线的垂直平分线组成的连续多边形组成，根据 n 个在平面上不重合种子点，把平面分成 n 个区域，使得每个区域内的点到它所在区域的种子点的距离比到其它区域种子点的距离近。

若一离散点集的点均处于一般位置，则德老内三角化就对应倒沃罗诺伊图的对偶。

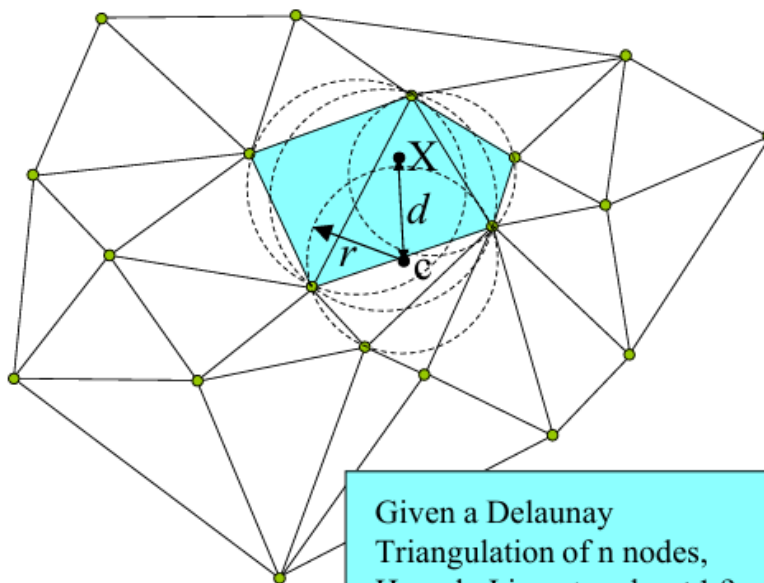


Given a Delaunay
Triangulation of n nodes,
How do I insert node $n+1$?

Lawson Algorithm

- Locate triangle containing X
- Subdivide triangle
- Recursively check adjoining triangles to ensure empty-circle property. Flip diagonal if needed
- (Lawson, 77)

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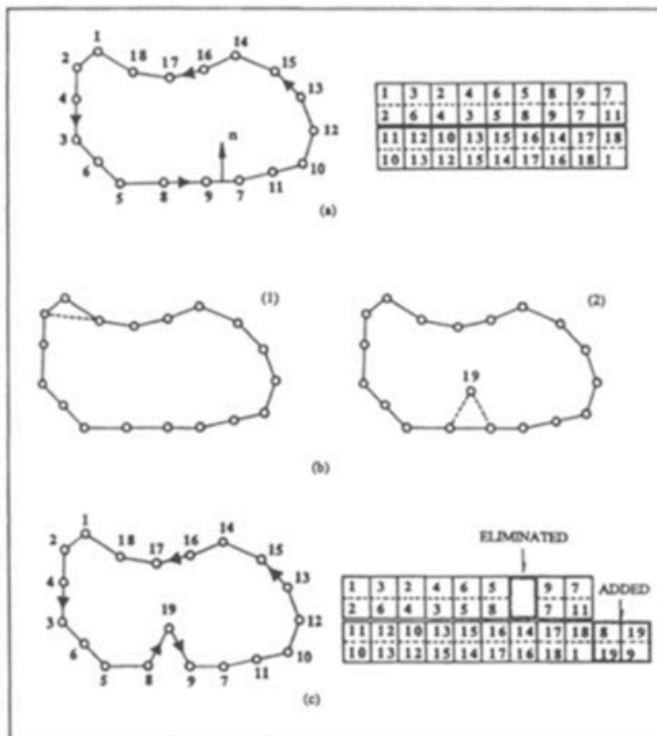


Given a Delaunay
Triangulation of n nodes,
How do I insert node $n+1$?

Bowyer-Watson Algorithm

- Locate triangle that contains the point
- Search for all triangles whose circumcircle contain the point ($d < r$)
- Delete the triangles (creating a void in the mesh)
- Form new triangles from the new point and the void boundary
- (Watson, 81)

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1. discretise the boundaries of the physical domain.
2. Generate a list of edges (faces in 3D) which represent the front.
3. Select the first edge (face) of the list and place a new point in the normal direction above the centre of the front edge (face).
4. Define a circle (sphere) with radius r centred at the point.
5. Determine all points which are located within the circle (sphere).
6. If there are no intersected points, generate a new element with the point. Otherwise, order the intersected points with respect to their distance to the point. Form elements with the points and accept the first one which does not intersect any other element and satisfies given quality measure(s).
7. Delete the current front edge (face) and add the newly formed edges (faces) to the list. Sort the list again.
8. Continue with step 3 until the list is empty.
9. Check the quality of the grid and/or swap edges if required.